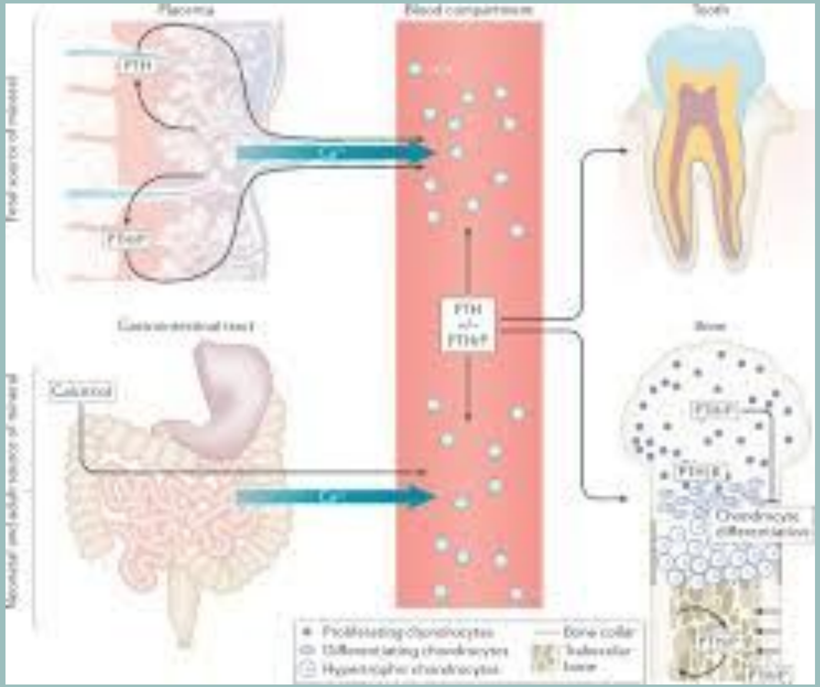


ROLE OF MINERALS IN TEETH AND BONES

DR. WAJEEHA SHAMS ANSARI

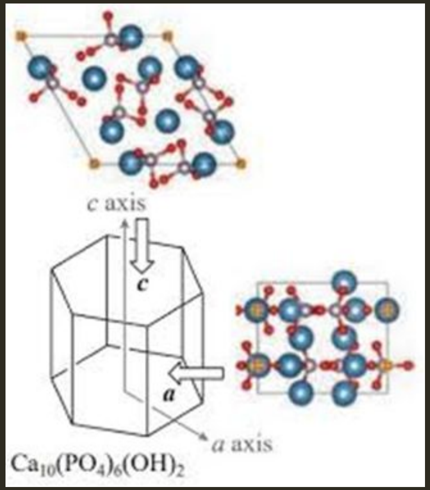
INTRODUCTION TO MINERALIZED TISSUES

- **Definition:** Mineralized tissues (bone, dentin, enamel) are composites of an organic matrix and an inorganic mineral phase.

- **Biochemical Goal:** Providing mechanical strength, structural support, and acid resistance.

- **Primary Mineral:** Hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$.

- **Critical pH of Enamel (Hydroxyapatite): 5.5**
- **Critical pH of Enamel (Fluorapatite): 4.5**



BIOCHEMISTRY OF HYDROXYAPATITE

- **Structure:** A stable, biocompatible calcium phosphate lattice (hexagonal crystal system).

- **Ionic Components:**
- **Calcium (Ca^{2+}):** Structural foundation.
- **Phosphate (PO_4^{3-}):** Critical for lattice stability.
- **Hydroxyl (OH^-):** Occupies the central channel of the crystal lattice.

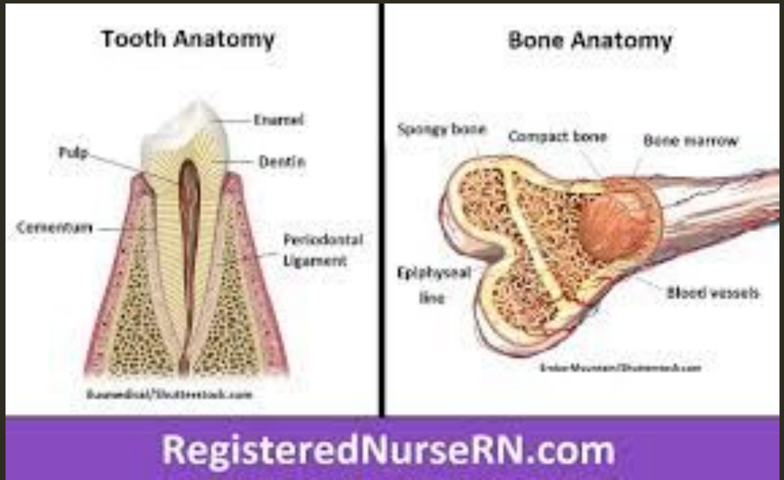
- **Substitution:** The OH^- group can be replaced by Fluoride (F^-) to form **Fluorapatite**, which is significantly more resistant to acid dissolution.

- **Difference between Enamel and Bone**: Enamel is an epithelial-derived, non-living, avascular tissue that cannot repair itself, whereas bone is a connective tissue that is constantly remodeling and vascularized.

Mineral / Ion	Primary Function in Hard Tissue	Clinical/Biochemical Significance
Calcium (Ca2+)	Structural backbone of Hydroxyapatite	Maintains serum homeostasis; essential for mineralization
Phosphate (PO_4 3-)	Crystal lattice formation	Buffering capacity in saliva/fluids
Fluoride (F-)	Crystal stability; acid resistance	Replaces OH^- to form Fluorapatite (FAP)
Magnesium (Mg2+)	Crystal size regulation	Influence on crystal growth and solubility
Carbonate (CO_3 2-)	Ionic substitution	Increases solubility; susceptible to acid erosion

MINERALIZATION: TEETH (ENAMEL & DENTIN)

- **<u>Enamel Composition</u>:** ~ 96% mineral, ~ 1% organic, ~ 3% water.
- **<u>Dentin Composition</u>:** ~ 70% mineral, ~ 20% organic (mostly Type I collagen), ~ 10% water.
- **<u>Process</u>:** Amelogenesis (enamel) is a unique process where ameloblasts deposit a mineral-heavy matrix without a collagen scaffold. Dentinogenesis relies on odontoblasts secreting a collagenous matrix that acts as a scaffold for mineral nucleation.



MINERALIZATION: BONE

- **<u>Matrix</u>:** Predominantly Type I collagen, serving as a nucleating site for crystals.
- **<u>Remodeling</u>:** A dynamic cycle of formation (osteoblasts) and resorption (osteoclasts).
- **<u>Function</u>:** Bone acts as the body's primary reservoir for calcium and phosphate, regulated by systemic hormonal demand.

HORMONAL & DIETARY REGULATION

- **Calcium Homeostasis:** Controlled by the Parathyroid Hormone (PTH) and Vitamin D axis.
- **Vitamin D:** Increases intestinal absorption of calcium and phosphate; essential for healthy mineralization.
- **PTH:** Stimulates bone resorption to maintain serum calcium levels during deficiency.
- **Clinical Correlation:** Deficiencies lead to conditions like rickets/osteomalacia (bone) or increased caries risk (teeth).

TRACE ELEMENTS & CLINICAL BIOCHEMISTRY

- **Magnesium (Mg^{2+}):** Regulates crystal size; deficiency can lead to brittle bone.

- **Fluoride (F^-):** Enhances enamel resistance by reducing solubility product (K_{sp}) of the mineral phase.

- **Carbonate (CO_3^{2-}):** Often substitutes for phosphate; increases crystal solubility (making tissue more prone to acid attack).

Tissue Component	Enamel (Mature)	Dentin	Bone
Inorganic Mineral (%)	~96%	~70%	~60–65%
Organic Matrix (%)	~1%	~20%	~25%
Water (%)	~3%	~10%	~10–15%
Primary Protein	Amelogenins	Type I Collagen	Type I Collagen
Living Tissue?	No	Yes (Odontoblasts)	Yes (Osteocytes)

SUMMARY

- Hydroxyapatite is the fundamental inorganic unit of teeth and bones.
- Mineralization is highly regulated, cell-mediated, and matrix-dependent.
- Dietary mineral intake and hormonal signaling are vital for long-term hard tissue structural integrity.


THANK YOU